Practice Set 5 Paper 1 – Basic Sciences in Urology

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Practice paper compiled from the question bank.

Q1. Discuss anatomy of urethra. Classify its congenital anomalies and discuss management of epispadias.

Type: LAQ (25 marks) (male urethral anatomy also answered in 2019-Summer Paper 1, Q1; isolated epispadias in 2022-Summer Paper 3, Q3f)

Anatomy of the male urethra

- About 18–20 cm long, 8–9 mm wide. **Posterior urethra:** bladder neck to **perineal membrane**; **anterior urethra:** perineal membrane to meatus, inside the corpus spongiosum.

Segment	Features	Lining
Prostatic	Urethral crest; **verumontanum** with utricle (Müllerian remnant) and ejaculatory ducts; angle divides preprostatic and prostatic parts	Urothelium
Membranous	2–2.5 cm; shortest and narrowest; striated sphincter	Pseudostratified columnar
Bulbar	Widens as the bulb; **Cowper ducts at 3 and 9 o'clock**; lies eccentric, nearer the dorsum	Pseudostratified columnar
Penile	About 15 cm; uniform calibre; glands of Littré	Pseudostratified columnar
Fossa navicularis	Glanular; lacuna magna on the roof	Stratified squamous

- **Sphincters:** (1) **internal (preprostatic) smooth-muscle sphincter**, autonomic, gives continence at rest; (2) **external striated rhabdosphincter**, horseshoe-shaped, supplied by the pudendal nerve (mainly S3, some S2; Onuf nucleus).
- **Arteries:** internal pudendal → common penile artery → bulbourethral branches; retrograde flow through the glans from the dorsal arteries. This **dual supply** allows bulbar transection in urethroplasty.
- **Veins** to the pudendal and prostatic plexuses; **lymph** to internal and external iliac nodes.
- **Female urethra:** about 4 cm; smooth muscle plus striated sphincter over the distal two-thirds; **Skene glands** (prostate homologue) near the meatus; oestrogen-dependent mucosal seal.

Classification of congenital urethral anomalies

Group	Anomalies
Abnormal meatal position	**Hypospadias** (ventral); **epispadias** (dorsal)
Obstructive	**PUV (posterior urethral valves)**, Young types I–III; **anterior urethral valves** (with diverticulum); urethral atresia or hypoplasia (congenital stricture); congenital meatal stenosis
Dilatation	**Megalourethra**: scaphoid (spongiosum deficient) or fusiform (all erectile tissue deficient, more severe); anterior urethral diverticulum; Cowper duct cyst (syringocele); enlarged prostatic utricle
Duplication	**Effmann classification**: type I blind incomplete; type II complete (IIA2 "Y-type"); type III with caudal duplication. The ventral channel is the functional one
Accessory openings	Congenital urethrocutaneous fistula; parameatal urethral cyst
Female	Epispadias; urethral prolapse; persistent urogenital sinus

Epispadias: definition and classification

- A **dorsal urethral wall defect**, replaced by a broad dorsal mucosal strip (urethral plate); part of the **exstrophy–epispadias complex**, mostly seen with exstrophy.
- Embryology: an overdeveloped cloacal membrane blocks mesenchymal ingrowth; genital tubercles fuse caudal to it.

	Male	Female
Incidence	About 1 in 117,000	About 1 in 484,000
Types	Glanular, penile, **penopubic** (about 70%, incontinent)	**Davis:** vesicular (patulous meatus), subsymphyseal, retrosymphyseal (incontinent)
Genitalia	Short, upturned penis with **dorsal chordee**	**Bifid clitoris**, flat glabrous mons
VUR (vesicoureteric reflux)	30–40%	30–75%

- The more proximal the meatus, the worse the continence (glanular never incontinent). Pubic diastasis is less than in exstrophy.

Evaluation

- Renal ultrasound; VCUG (voiding cystourethrogram) for reflux; cystoscopy and **bladder capacity** under anaesthesia.
- **Key decision:** use the exstrophy pathway (osteotomy and closure) if pubic diastasis >3 cm, poor intrasymphyseal bar, absent sphincter tissue or tiny bladder.

Management

Aims: continence with safe upper tracts, and a straight penis of adequate length that works for intercourse.

- **Preoperative testosterone** (2 mg/kg intramuscular, 5 and 2 weeks before).

Stage 1: urethroplasty and penile reconstruction (6–12 months)

1. Release dorsal chordee: divide suspensory ligaments; free corpora from the inferior pubic rami.
2. **Modified Cantwell–Ransley:** an 18-mm plate strip is outlined; reverse MAGPI (meatal advancement and glanuloplasty) brings the meatus to the tip; the plate is tubularised over an 8-Fr stent and moved **ventral to the corpora**, which are rotated medially over it (diamond corporotomies if curvature persists).
3. Glans wings closed; skin flaps resurface the dorsum; base Z-plasty; stent 10–12 days.

- **Mitchell–Bägli complete penile disassembly:** reliable chordee correction, but often leaves hypospadias.
- Urethroplasty adds outlet resistance, so the bladder grows before stage 2.

Stage 2: bladder neck reconstruction (4–5 years, capacity about 100 mL)

- **Young–Dees–Leadbetter** repair with Marshall–Marchetti–Krantz suspension and **ureteric reimplantation**. Osteotomy is rarely needed.
- **Good capacity at presentation:** single-stage repair is possible.
- **Female:** urethroplasty and genitoplasty (clitoral halves and labia joined, mons reconstructed) at 6–12 months; bladder neck repair at 4–5 years.

Outcomes and complications

- Continence after bladder neck reconstruction: **males 47–82%, females 70–87%**; capacity is the main predictor.

- Early fistula about 1 in 6 (penile base); stricture rare; neourethra catheterisable; erections and paternity preserved.
- Better than exstrophy: the bladder was never exposed or scarred. Failures: injectables, augmentation.

Diagram to draw: sagittal male urethra with segments; Cantwell–Ransley with plate transposed under the corpora.

> **Key points to remember:**
> - Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock.
> - Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
> - About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
> - Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

Q2. Write in detail about physiology of erection. How will you evaluate a diabetic with ED?

Type: LAQ (25 marks)

Functional anatomy

- **Corpora cavernosa:** sinusoids lined by endothelium, separated by **trabecular smooth muscle**. Flaccid tone depends on smooth-muscle contraction.
- **Tunica albuginea:** inner circular and outer longitudinal layers. **Emissary veins** pass obliquely between them and are compressed when the corpora expand: the **veno-occlusive mechanism**.
- **Arteries:** internal pudendal → common penile → **cavernous (helicine branches)**, dorsal and bulbourethral arteries. Accessory pudendal arteries in about one-third.
- Spongiosum and glans have a thin or absent tunica, so no veno-occlusion; they stay low-pressure.

Haemodynamic phases

Phase	Events	Intracavernous pressure
Flaccid	Smooth muscle contracted; low inflow	Low
Latent (filling)	Arterial dilation; **inflow peaks**	Barely rises
Tumescence	Sinusoids fill; venous outflow compressed	Rising
Full erection	Maximal veno-occlusion	About 100 mmHg
Rigid	**Ischiocavernosus** contraction (pudendal nerve)	Several hundred mmHg
Detumescence	Initial, slow, fast phases	Falls to baseline

Molecular mechanism

- **Relaxation (erection):** NO (nitric oxide) from **nNOS (neuronal nitric oxide synthase)** in cavernous nerves initiates erection; endothelial NOS (eNOS) sustains it.
  - NO → soluble guanylyl cyclase → **cGMP (cyclic guanosine monophosphate)** → protein kinase G → K^+ channels open, Ca^{2+} channels close, Ca^{2+} sequestered → **low cytosolic Ca^{2+}** → relaxation.
  - Second route: PGE1 (prostaglandin E1), VIP (vasoactive intestinal peptide) → **cAMP (cyclic adenosine monophosphate)** → protein kinase A.
  - **PDE5 (phosphodiesterase type 5)** degrades cGMP and ends relaxation; PDE5 inhibitors block it.
- **Contraction (flaccidity): noradrenaline (α1)**, endothelin-1, angiotensin II → raised Ca^{2+} → calmodulin → myosin light-chain kinase → contraction. **Rho-kinase** inhibits myosin phosphatase and maintains tone.
- **Detumescence:** NO release stops, PDE5 degrades cGMP and sympathetic discharge contracts the smooth muscle, so emissary veins reopen.
- Gap junctions (connexin-43) synchronise the smooth muscle. **Also know:** carbon monoxide and hydrogen sulfide also raise cGMP; acetylcholine acts indirectly through endothelial NO (atropine does not block erection).

Neural control

Pathway	Origin	Role
Parasympathetic	**S2–S4** → pelvic plexus → cavernous nerves	Erection

Pathway	Origin	Role
Sympathetic	**T11–L2** → hypogastric nerves	Detumescence, emission
Somatic	Pudendal nerve (Onuf nucleus)	Penile sensation; ischiocavernosus rigidity

- **Three types: psychogenic** (brain, chiefly by reducing sympathetic tone), **reflexogenic** (genital touch, sacral arc; preserved in upper cord injury), **nocturnal** (REM [rapid eye movement] sleep).
- **Central:** MPOA (medial preoptic area) and **PVN (paraventricular nucleus)**. Dopamine and oxytocin promote; serotonin, prolactin, GABA (γ-aminobutyric acid) and opioids inhibit.

Evaluation of a diabetic with ED (erectile dysfunction)

Why diabetics: ED is commoner, earlier and more severe in DM (diabetes mellitus). Mechanisms: endothelial NO loss, autonomic neuropathy, arteriopathy, smooth-muscle loss and fibrosis (veno-occlusive failure), AGEs (advanced glycation end products) and low testosterone. ED is a **sentinel of cardiovascular disease**.

1. History

- Onset, duration (≥3 months), severity, morning erections, libido, ejaculation; partner.
- **IIEF-5 (International Index of Erectile Function, SHIM):** 5–7 severe, 8–11 moderate, 12–16 mild-to-moderate, 17–21 mild, 22–25 no ED.
- Diabetes: type, duration, **glycaemic control**, complications (retinopathy, nephropathy, neuropathy, foot disease), autonomic symptoms (postural giddiness, gastroparesis, bladder dysfunction).
- Cardiovascular: angina, exercise tolerance, smoking, hypertension, dyslipidaemia, nitrates.
- Drugs: antihypertensives, antidepressants. Depression screen.

2. Examination

- BMI, waist, blood pressure, peripheral pulses.
- Genitalia: Peyronie plaques, testicular size, gynaecomastia.
- Neurological: genital sensation, **bulbocavernosus reflex**, foot sensation. Penile neuropathy can exist without foot neuropathy.

3. Laboratory

- **HbA1c (glycated haemoglobin)** and fasting glucose, **lipid profile**, creatinine/eGFR (estimated glomerular filtration rate), urine albumin.
- **Morning total testosterone**; if low, repeat with free testosterone, LH (luteinising hormone), prolactin. TSH (thyroid-stimulating hormone) if indicated.

4. Cardiovascular risk stratification (Princeton Consensus)

- **High risk** (unstable angina, recent infarction, uncontrolled hypertension, some arrhythmias): defer sexual activity and treatment; cardiology referral. **Indeterminate:** cardiac assessment (e.g. stress test), then restratify. **Low:** start treatment.

5. Specialised tests (selected cases)

- **CIS (combined intracavernous injection and stimulation):** alprostadil 10–20 µg; a rigid erection excludes major vascular disease.
- **Penile duplex Doppler:** PSV (peak systolic velocity) **<25 cm/s = arterial insufficiency** (>35 normal); EDV (end-diastolic velocity) **>5 cm/s** and RI (resistive index) **<0.75 = veno-occlusive dysfunction**.
- **NPTR (nocturnal penile tumescence and rigidity), RigiScan:** separates psychogenic from organic ED.
- **Neurophysiological:** bulbocavernosus reflex latency, dorsal nerve conduction velocity, biothesiometry.

- DICC (dynamic infusion cavernosometry and cavernosography) and angiography: rarely, as diffuse diabetic vascular disease is not suited to revascularisation.

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Diabetic with ED → history + IIEF-5 + examination
   → HbA1c, lipids, testosterone, renal function
   → Princeton cardiac risk → treat risk factors
   → PDE5 inhibitor trial → failure → duplex/CIS → ICI or prosthesis
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Diagram to draw: NO–cGMP pathway in the cavernous smooth-muscle cell.

Key points to remember:

- nNOS initiates, eNOS maintains; PDE5 ends erection.
- Rigid phase: ischiocavernosus, suprasystolic pressure.
- Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
- Diabetic ED: check HbA1c, testosterone and cardiac risk.

Q3a. Tear drop bladder

Type: SAQ (10 marks)

Definition

- A **tear-drop (pear-shaped or inverted-pear) bladder** is a bladder that is symmetrically **compressed from both sides** and elongated upwards on cystogram, IVU (intravenous urogram) or CT (computed tomography).
- It is a radiological sign of a **bilateral perivesical mass or collection**, not a disease itself.

Causes

Group	Causes
Collections	**Pelvic haematoma** (pelvic fracture: commonest in trauma); urinoma (extraperitoneal leak); bilateral lymphocele (after lymphadenectomy or transplant); pelvic abscess
Fat	**Pelvic lipomatosis**
Muscle	**Iliopsoas hypertrophy** (muscular young men)
Vascular	Dilated pelvic venous collaterals (IVC [inferior vena cava] occlusion); bilateral iliac artery aneurysms
Nodes and tumour	Pelvic lymphadenopathy (lymphoma, metastases); pelvic retroperitoneal fibrosis; pelvic tumours

Pelvic lipomatosis (the classic cause)

- Overgrowth of benign, non-encapsulated fat in the perivesical and perirectal spaces; mainly in men, more in Black men.
- **Presentation:** LUTS (lower urinary tract symptoms), suprapubic pain, constipation; often incidental.
- **Imaging:**
  - Plain film: pelvic lucency. IVU: elevated, elongated pear-shaped bladder; **medial deviation of both ureters**; hydronephrosis.
  - Barium enema: straightened, elongated rectosigmoid.
  - **CT or MRI (magnetic resonance imaging):** fat density is diagnostic.
- **Cystoscopy:** high bladder neck, difficult passage; often **cystitis glandularis**, which can progress to adenocarcinoma. Biopsy and surveillance are needed.
- **Management:** observe if no obstruction. Upper-tract obstruction → stent or nephrostomy, ureteric reimplantation or urinary diversion. Weight loss rarely helps; surgical excision of fat is difficult.

Evaluation of a tear-drop bladder

- History: trauma, pelvic surgery or transplant, malignancy.
- **CT** shows the cause (blood, fat, fluid, muscle, nodes, vessels). Ultrasound for collections; aspiration (creatinine for urinoma, lymph for lymphocele).

Management

- Treat the cause: pelvic haematoma usually observed; lymphocele drained or marsupialised (laparoscopic fenestration); urinoma drained and leak repaired.

Diagram to draw: cystogram outline of a bladder squeezed into a tear drop by bilateral pelvic haematoma.

Key points to remember:

- Tear-drop bladder = bilateral extrinsic compression.
- Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
- Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

Q3b. TESA

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 2, Q3c)

Definition

- **TESA (testicular sperm aspiration):** percutaneous needle aspiration of seminiferous tubules under local anaesthesia to retrieve sperm for **ICSI (intracytoplasmic sperm injection)**, which needs one sperm per oocyte.

Indications

- **OA (obstructive azoospermia) with normal spermatogenesis:** CBAVD (congenital bilateral absence of the vas deferens), unreconstructable obstruction, failed vasectomy reversal, or failed epididymal aspiration.
- Anejaculation (spinal cord injury) after failed vibratory stimulation or electroejaculation.
- **NOA (non-obstructive azoospermia):** poor yield; **micro-TESE (microdissection testicular sperm extraction)** preferred.

Technique

1. Spermatic cord block with local anaesthetic.
2. Testis fixed; **23-G needle** (or angiocatheter) on a high-suction syringe.
3. Multiple passes in different directions; tubules drawn into the needle (often 10–20 passes needed).
4. Immediate wet examination for sperm; repeat until sperm are found. **Cryopreserve** spare sperm.

Comparison of retrieval methods

Technique	Site	Best for	Notes
PESA (percutaneous epididymal sperm aspiration)	Caput epididymis	OA	Motile but few sperm; may obstruct epididymis
MESA (microsurgical epididymal sperm aspiration)	Single tubule at 16–25×	OA	Most motile sperm; many frozen vials
TESA	Testis, needle	OA	Least invasive; few, immotile sperm
Percutaneous biopsy	14-G biopsy gun	OA	Core of tissue; blind
TESE / micro-TESE	Open, larger tubules under microscope	**NOA**	Most sperm, least tissue

Advantages and limitations

- **Advantages:** office procedure, no microsurgery, cheap, repeatable.
- **Limitations:** few tubules with poor architecture (not diagnostic histology); small numbers of immature, immotile sperm; often too few to freeze; blind.

Complications

- Haematoma (commonest), pain, testicular vascular injury, infection, epididymal injury.

Outcome

- OA: retrieval almost always succeeds; ICSI is essential for testicular sperm.
- NOA: fine-needle aspiration yields far fewer sperm than micro-TESE.

- Before ICSI: karyotype, Y-microdeletion and CFTR (cystic fibrosis transmembrane conductance regulator) testing as indicated.

Diagram to draw: testis with 23-G needle on a suction syringe and multiple passes.

> **Key points to remember:**
> - TESA: 23-G needle, local anaesthesia, OA with normal spermatogenesis.
> - NOA: micro-TESE best; TESA yield poor.
> - Testicular sperm always need ICSI; cryopreserve extras.

Q3c. ESBL producing enterobacterial UTI.

Type: SAQ (10 marks) (also answered in 2019-Summer Paper 1, Q3f)

Definition

- **ESBLs (extended-spectrum β-lactamases): plasmid-encoded** enzymes that hydrolyse penicillins, **third-generation cephalosporins** and aztreonam. They are **inhibited by clavulanate** and spare **carbapenems** and cephamycins.
- Mainly *E. coli* and *Klebsiella* (Enterobacterales); first reported 1983.

Mechanism and spread

- Older types **TEM** and **SHV**; **CTX-M** now dominates. Community spread is linked to *E. coli* **ST131**, whose hallmark is fluoroquinolone resistance.
- Plasmids carry co-resistance to **fluoroquinolones, aminoglycosides and TMP-SMX (trimethoprim–sulfamethoxazole)**, so oral options are few.
- India has one of the highest prevalence rates.

Risk factors

- Prior antibiotics (cephalosporins, quinolones); recent hospital stay; **indwelling catheter** and instrumentation; recurrent UTI (urinary tract infection); diabetes; nursing-home residence; travel; transrectal prostate biopsy.

Detection

- Screening: reduced susceptibility to ceftriaxone, cefotaxime or ceftazidime.
- **Phenotypic confirmation:** clavulanate synergy (combination disc or double-disc test); automated systems.
- Genotypic PCR (polymerase chain reaction) for CTX-M.

Treatment

Infection	Options
Cystitis	**Nitrofurantoin** 5 days; **fosfomycin** 3 g single dose; **pivmecillinam**; TMP-SMX or quinolone only if susceptible
Pyelonephritis, complicated UTI	Quinolone or TMP-SMX if susceptible; otherwise **aminoglycoside**
Urosepsis, bacteraemia	**Carbapenem** (meropenem, ertapenem), then oral step-down by culture

- **MERINO trial (2018):** in ceftriaxone-resistant *E. coli*/*Klebsiella* bacteraemia, **piperacillin–tazobactam** had higher 30-day mortality than meropenem (12.3% vs 3.7%).
- Ceftazidime–avibactam and ceftolozane–tazobactam are **reserved** for carbapenem-resistant organisms.
- **Source control:** change or remove catheters; drain obstruction.
- Treat **asymptomatic bacteriuria** only before urological procedures or in pregnancy.
- **Febrile prostatitis after transrectal biopsy:** carbapenem is favoured.

Prevention

- **Antibiotic stewardship:** culture before treatment; narrowest spectrum; shortest course.
- Hand hygiene, contact precautions, catheter care; rectal swab culture before biopsy, or transperineal biopsy.

Also know: AmpC (not inhibited by clavulanate); carbapenemases KPC (*Klebsiella pneumoniae* carbapenemase) and NDM (New Delhi metallo-β-lactamase).

Diagram to draw: flowchart from culture result to oral or intravenous choice.

Key points to remember:

- Plasmid CTX-M; resists third-generation cephalosporins; clavulanate inhibits.
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: piperacillin–tazobactam inferior.

Q3d. Flow cytometry

Type: SAQ (10 marks)

Definition

- **Flow cytometry** measures physical and fluorescent properties of **single cells in suspension** flowing one at a time through a laser beam, thousands per second.

Principle

1. **Sample:** fresh cells, urine, bladder washings, or nuclei from paraffin blocks.
2. **Staining:** fluorochrome-labelled antibodies to CD (cluster of differentiation) antigens, or **DNA-binding dyes** (propidium iodide, DAPI [4′,6-diamidino-2-phenylindole]).
3. **Hydrodynamic focusing** lines the cells up in single file.
4. **Laser interrogation:**
    - **Forward scatter** = cell size.
    - **Side scatter** = internal complexity (granularity).
    - **Fluorescence** = amount of bound antibody or DNA.
5. Photomultiplier detectors → histograms and dot plots; **gating** selects populations.

- **FACS (fluorescence-activated cell sorting)** physically separates selected cells.

DNA content analysis

- **DNA index** = tumour G0/G1 peak ÷ normal diploid peak. DI 1.0 = diploid; about 2.0 = tetraploid; any other = **aneuploid**.
- **S-phase fraction** estimates proliferation.
- Aneuploidy and high S-phase generally mean higher grade and worse prognosis.

Applications in urology

Area	Use
Bladder cancer	Ploidy of bladder washings: aneuploidy with high grade, CIS (carcinoma in situ), recurrence; replaced by **UroVysion FISH (fluorescence in situ hybridisation)**
Upper-tract urothelial cancer	Aneuploidy → poor long-term survival
Prostate cancer	Aneuploid tumours progress more after treatment than diploid
Neuroblastoma	**Hyperdiploid favourable**; diploid or tetraploid adverse; part of COG (Children's Oncology Group) risk
Urinalysis	Automated urine analysers count RBC (red blood cells), WBC (white blood cells) and bacteria
Transplantation	**Flow cytometric crossmatch**: more sensitive than CDC (complement-dependent cytotoxicity) crossmatch for donor-specific antibody; Luminex single-antigen beads; CD3 count to guide antithymocyte globulin
Male infertility	Sperm DNA fragmentation (SCSA [sperm chromatin structure assay]); haploid cells in testicular aspirates
Lymphoma	Immunophenotyping of testicular or renal lymphoma

Advantages and limitations

- **Advantages:** rapid, objective, quantitative, many parameters per cell.
- **Limitations:** architecture lost; near-diploid aneuploidy missed; normal cells dilute tumour peaks; ploidy now adds little independent value.

Diagram to draw: DNA histogram with diploid G0/G1 peak, S-phase and aneuploid peak.

Key points to remember:

- Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
- DNA index 1.0 diploid; aneuploidy = worse prognosis.
- Flow crossmatch is more sensitive than CDC crossmatch.

Q3e. Vascular grafts in urology

Type: SAQ (10 marks) (also answered in 2021-Summer Paper 3, Q2 and 2015-Summer Paper 2, Q3e)

Introduction

- A vascular graft **replaces, bypasses, extends or widens** a vessel when primary repair would leave tension or narrowing. Patch widens a wall; interposition graft replaces a segment; bypass routes flow around a lesion.

Types

Type	Examples	Advantages	Disadvantages
Autologous vein	Saphenous, gonadal, donor external iliac vein	Resists infection; small vessels	Limited length
Autologous artery	Internal iliac, **Carrel aortic patch**	Best for renal artery	Limited supply
Xenograft	Bovine pericardium	Pliable patch	Patch only
Synthetic	**PTFE (polytetrafluoroethylene)**, Dacron	Large calibre, any length	Infection; venous grafts thrombose

Applications

1. **Renal transplantation:** multiple arteries (18–43%) joined by Carrel patch or bench reconstruction; **preserve the lower-pole artery** (supplies the ureter). Short right renal vein extended with donor IVC (inferior vena cava), external iliac or spiral gonadal vein.
2. **RCC (renal cell carcinoma) with IVC thrombus:** residual lumen <50% → **patch cavoplasty**; circumferential wall invasion → **PTFE replacement (16–20 mm)**, then lifelong aspirin.
3. **PC-RPLND (post-chemotherapy retroperitoneal lymph node dissection):** IVC narrowing >25% → patch or graft; IVC ligation tolerated. **Aorta never ligated**: PTFE or Dacron replacement (about 1%).
4. **Retroperitoneal sarcoma:** caval resection and graft; thrombosis 10–40%.
5. **Renovascular repair and autotransplantation:** ex vivo bench repair with internal iliac artery or saphenous vein.
6. **Dialysis access:** PTFE AV (arteriovenous) graft when no native fistula is possible.
7. **Penile:** Grayhack saphenous–cavernosal shunt for priapism.

Principles

- Autologous in contaminated fields and small vessels; synthetic for large, high-flow vessels.
- Venous grafts: use the widest graft. Heparin before clamping; tension-free spatulated anastomosis with monofilament polypropylene; **omental wrap** keeps synthetic grafts away from bowel and urine.

Complications

- Thrombosis, stenosis, infection (synthetic usually removed), graft–enteric fistula, **ureteroarterial fistula**.

Diagram to draw: IVC patch cavoplasty and PTFE caval graft.

Key points to remember:

- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- IVC: lumen <50% → patch; invaded wall → PTFE 16–20 mm.
- PC-RPLND: IVC may be ligated; aorta must be grafted.

Q3f. Hypokalemia

Type: SAQ (10 marks)

Definition and physiology

- **Hypokalaemia:** serum potassium **<3.5 mEq/L**; severe <2.5 mEq/L.
- About **98% of body potassium is intracellular**; the ratio sets the resting membrane potential.
- Two defences: **shift into cells** (insulin, β2-adrenergic stimulation, alkalosis) and **renal excretion** (distal nephron). Distal secretion rises with high urine flow, distal sodium delivery, **aldosterone** and alkalosis.

Causes

Mechanism	Causes
Low intake	Malnutrition
Shift into cells	Insulin, catecholamines (stress, β-agonists), alkalosis, refeeding, periodic paralysis
GI (gastrointestinal) loss	Diarrhoea, villous adenoma, **ureterosigmoidostomy and colonic diversions** (colon secretes potassium, ileum absorbs it)
Renal loss, high BP (blood pressure)	**Primary hyperaldosteronism (Conn)**, renin-secreting juxtaglomerular tumour, Cushing, Liddle, **abiraterone** (mineralocorticoid excess)
Renal loss, normal BP	Loop and thiazide diuretics, **postobstructive diuresis**, osmotic diuresis, **RTA (renal tubular acidosis) types 1 and 2**, Bartter, Gitelman, hypomagnesaemia, amphotericin, cisplatin

- **Vomiting:** potassium is lost mainly in urine (alkalosis → kaliuresis), not in gastric juice.

Clinical features

- Muscle weakness, cramps, ileus, **flaccid paralysis**, rhabdomyolysis.
- Arrhythmias; potentiates digoxin toxicity.
- Polyuria (nephrogenic diabetes insipidus); **hypocitraturia** → calcium stones.
- **ECG (electrocardiogram):** flat T wave, **prominent U wave**, ST depression, prolonged QU; ventricular arrhythmias.

Evaluation

- Repeat potassium; magnesium, bicarbonate, glucose.
- **24-h urine potassium:** <20 mmol/day suggests extrarenal loss; higher suggests renal loss.
- Renal loss: check blood pressure and acid–base; **aldosterone–renin ratio** if hypertensive (only a minority of Conn patients are hypokalaemic).

Management

- Treat the cause; stop offending drugs.
- **Potassium chloride** preferred (potassium citrate if acidotic, phosphate if hypophosphataemic). **Oral** 20–40 mEq repeated when possible.
- **Intravenous** for severe or symptomatic cases: usually ≤10 mEq/h peripherally; faster only centrally with ECG monitoring. Never as a bolus.
- **Correct magnesium**, or potassium will not correct.
- **Urinary diversion:** replace potassium **before or with bicarbonate**; correcting acidosis alone drives potassium into cells → paralysis.

- Correct before abiraterone and adrenalectomy for Conn syndrome (spironolactone preoperatively).

Diagram to draw: ECG strip showing flat T and U wave.

> **Key points to remember:**
> - Shift or loss; urine potassium separates renal from extrarenal.
> - ECG: flat T, U wave.
> - Diversion acidosis: give potassium with or before bicarbonate.
> - Correct magnesium first.

Rapid Revision Sheet – Practice Set 5 Paper 1

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

Q1. Discuss anatomy of urethra. Classify its congenital anomalies and discuss management of epispadias. (LAQ)

- Bulbar urethra eccentric (dorsal); Cowper ducts 3 and 9 o'clock.
- Megalourethra: fusiform worse than scaphoid; duplication: ventral channel functional.
- About 70% of male epispadias are penopubic; diastasis >3 cm → exstrophy pathway.
- Cantwell–Ransley at 6–12 months; YDL (Young–Dees–Leadbetter) at about 100 mL.

Q2. Write in detail about physiology of erection. How will you evaluate a diabetic with ED? (LAQ)

- nNOS initiates, eNOS maintains; PDE5 ends erection.
- Rigid phase: ischiocavernosus, suprasystolic pressure.
- Duplex: PSV <25 cm/s arterial; EDV >5 cm/s, RI <0.75 venous leak.
- Diabetic ED: check HbA1c, testosterone and cardiac risk.

Q3a. Tear drop bladder (SAQ)

- Tear-drop bladder = bilateral extrinsic compression.
- Trauma: pelvic haematoma; non-trauma: pelvic lipomatosis.
- Lipomatosis: fat on CT, medially deviated ureters, cystitis glandularis.

Q3b. TESA (SAQ)

- TESA: 23-G needle, local anaesthesia, OA with normal spermatogenesis.
- NOA: micro-TESE best; TESA yield poor.
- Testicular sperm always need ICSI; cryopreserve extras.

Q3c. ESBL producing enterobacterial UTI. (SAQ)

- Plasmid CTX-M; resists third-generation cephalosporins; clavulanate inhibits.
- Oral: nitrofurantoin, fosfomycin, pivmecillinam.
- Serious infection: carbapenem; MERINO: piperacillin–tazobactam inferior.

Q3d. Flow cytometry (SAQ)

- Forward scatter = size; side scatter = granularity; fluorescence = marker or DNA.
- DNA index 1.0 diploid; aneuploidy = worse prognosis.
- Flow crossmatch is more sensitive than CDC crossmatch.

Q3e. Vascular grafts in urology (SAQ)

- Autologous for small or contaminated fields; PTFE or Dacron for large vessels.
- IVC: lumen <50% → patch; invaded wall → PTFE 16–20 mm.
- PC-RPLND: IVC may be ligated; aorta must be grafted.

Q3f. Hypokalemia (SAQ)

- Shift or loss; urine potassium separates renal from extrarenal.
- ECG: flat T, U wave.
- Diversion acidosis: give potassium with or before bicarbonate.
- Correct magnesium first.